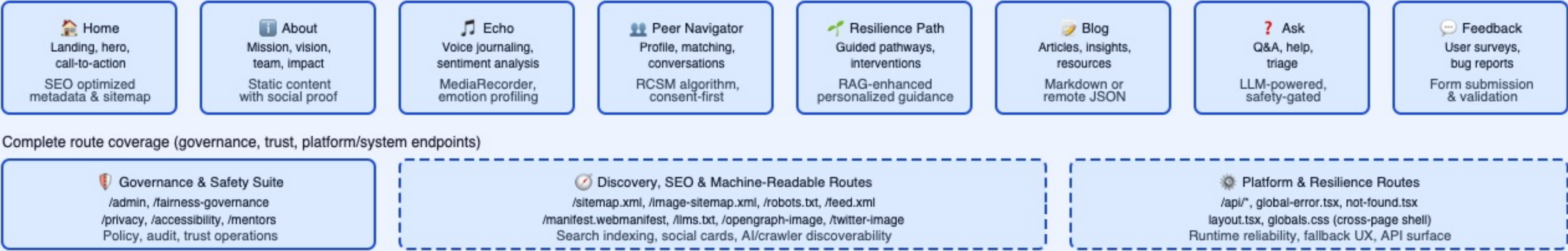


Aether Architecture: Layered System Design

Privacy-First Student Resilience Platform | Production Monorepo

LAYER 1: USER INTERFACE & PAGES

Public-facing pages, entry points, and primary user interactions. Next.js App Router with Server & Client components.



LAYER 2: CORE FEATURES & AI SERVICES

Business logic, algorithms, and intelligent systems. React components + TypeScript utilities.



LAYER 3: DATA, STATE & LOCAL STORAGE

Data persistence, caching, and state management. Browser APIs + future backend integration.



LAYER 4: BACKEND SERVICES & API

Node.js HTTP services, endpoints, and business logic. Ready for scaling to microservices.



LAYER 5: INFRASTRUCTURE & EXTERNAL SERVICES

Deployment, CDN, external APIs, and third-party integrations. Production-ready infrastructure.



Key Architectural Principles

- ✓ Privacy-First: Client-side processing where possible, minimal data collection, GDPR-aware
- ✓ Scalable Monorepo: Workspaces for apps & packages, independent CI/CD, shared types
- ✓ Modular AI: RAG for resilience guidance, RCSM for peer matching, optional LLM services
- ✓ Safety-Gated: Consent flows, content review, fairness audits, crisis escalation paths
- ✓ Production-Ready: Type-safe (TypeScript), tested (Jest), performant (Next.js 14), observable (analytics)
- ✓ Cloud-Native: Docker-ready backend, Vercel edge functions, async jobs & workers ready

- Primary user-flow route (core experience)
- System/platform route (SEO, API, error-handling, infrastructure)
- Cross-layer data flow direction